



Four clinical ECG sources

PTB-XL | Georgia | Chapman | Ningbo

77,333 records -> 66,379 retained, 13 harmonised labels

10 s ECG extraction



Native 500 Hz ECG

signal-quality characterisation

RQ1



Source signal characteristics

HF | baseline | mains
flat/clipped | QRS | RR

quality indicators

RQ3



Quality-performance association

within-source strata | prevalence matching
dAUPRC with 95% CI



Anti-aliased resampling to 100 Hz

FIR zero phase | input 12 x 1000

RQ2



External-source discrimination and selective reliability

leave-one-source-out | 6 encoders x 4 folds
AUPRC higher | E-AURC lower

LOSO predictions

trained encoders

RQ4



Embedded feasibility of the externally evaluated models

STM32F411 100 MHz | int8 PTQ
Flash | SRAM | latency | dAUPRC

discrimination

hardware metrics

RQ5



Accuracy-resource operating points

diagnostic value <-> deployment cost
macro AUPRC | Flash / SRAM / measured latency
which encoders are worth running on the part